

FIELD INSPECTION & TELEMETRY AUDIT REPORT

Document ID: IR-2026-0914-A | **Unit:** Machine 101 (ICU-101-A) | **Date:** Sep 14, 2026
Inspector: R. Sharma (Badge: ENG-4421) | **Location:** Machinery Bay 3

1. Recorded Field Telemetry

Parameter	Measured Value	Baseline Spec	Status
Core Temperature	84.2°C	60°C - 80°C	WARNING (>80°C)
Ventilation Fan	1240 RPM	1200 - 1400 RPM	COMPLIANT (ACTIVE)
Coolant Pressure	2.35 bar	2.0 - 3.5 bar	COMPLIANT
Coolant Level	68%	60% - 100%	COMPLIANT
Vibration Index	0.19 mm/s	< 0.30 mm/s	COMPLIANT

2. Physical & Visual Observations

- **Radiator Louvers & Intake Grill:** Visual inspection reveals moderate lint and industrial airborne dust accumulation covering approximately 20% to 30% of radiator surface area. Airflow intake partially restricted.
- **Cooling Fan Assembly:** Casing intact. Fan rotating smoothly at 1240 RPM. Zero audible bearing squeal or mechanical imbalance.
- **Coolant Fluid Lines:** Zero leaks, drips, or weeping connections along primary or secondary manifolds.
- **Thermocouple:** Thermocouple probe firmly seated; cross-verified against handheld IR probe at 84.0°C.

3. Inspector Evaluation & Recommendation

Thermal elevation (84.2°C) is directly caused by airflow impedance from 20-30% radiator dust accumulation. Fan speed (1240 RPM) and coolant level (68%) satisfy equipment specifications. Requesting Chief Maintenance Engineer prepare an official Approval Note for continued conditional operation subject to radiator fin cleaning within 48 hours.